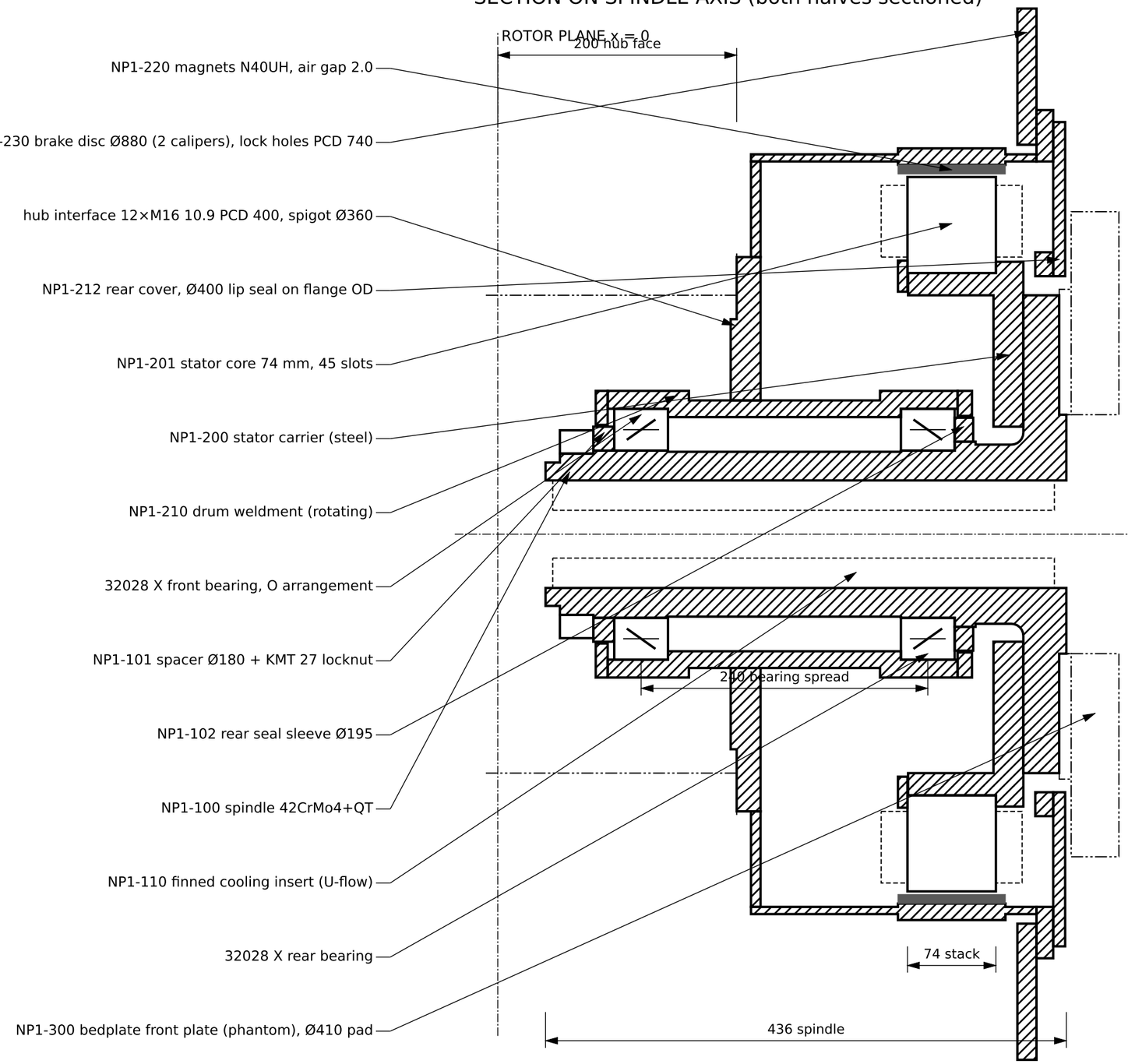
SECTION ON SPINDLE AXIS (both halves sectioned)



Item	Part	Qty	Material / type
1	NP1-100 spindle	1	42CrMo4+QT forged
2	NP1-101 spacer	1	42CrMo4, induction hard.
3	32028 X bearing	2	bought
4	KMT 27 locknut	1	bought
5	NP1-110 cooling insert	1	6082-T6 extrusion
6	NP1-200 stator carrier	1	S355J2+N
7	NP1-201 stator core	≈205 sheets	M270-35A bonded
8	NP1-202 clamp ring	1	1.4404
9	NP1-210 drum weldment	1	S355J2+N
10	NP1-211 seal covers	2	S355J2
11	NP1-212 rear cover	1	5083-H111
12	NP1-220 magnets	160	N40UH bought
13	NP1-230 brake disc	1	C45E+N

NOTES

1. x is measured rearward from the rotor plane along the spindle axis; the spindle axis is tilted 5° nose up (bedplate NP1-300).
2. Stationary: spindle, cones, stator carrier, stator, cooling insert, bedplate. Rotating: drum, cups, seal covers, rear cover, magnets, brake disc, hub.
3. Bearings: back-to-back (O) tapered roller pair, preload 2 to 4 kN (end play −0.02 to −0.05) set by the KMT 27 locknut, checked by rolling torque before the front seal cover is fitted (doc 14, 5.3; it now goes on after the nut).
4. Rev B (doc 17): disc Ø880/Ø652 slides on from the front, lock holes in the disc only, seal sleeve NP1-102 behind the rear cone, spacer Ø180, bedplate face on a Ø410 pad (rear cover runs 5 mm from the plate).
5. Air gap 2.0 mm nominal; eccentricity ≤ 0.20 mm after assembly, measured with 4 capacitive gap sensors (doc 15).
6. Sealed generator volume: front by the hub, rear by the Ø400 lip seal + labyrinth, desiccant breather on the rear cover.
7. Hub, pitch system, nacelle cover and slip ring are not part of P1 bench article (dummy hub NP1-600 instead).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

GENERAL ARRANGEMENT · DRIVETRAIN SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-000	B · 2026-11-03	1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see parts list	≈ 320 kg (drivetrain, no hub)	Claude / not checked